

SAHIL BODKE

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SUMMARY

Software Engineer with experience building Python APIs, cloud-native data pipelines on AWS and Azure, and AI/ML workflows. Background in containerization, CI/CD, and distributed systems through multiple co-ops and full-time roles

EDUCATION

Northeastern University (Khoury College of Computer Sciences), Boston, MA

Sept 2022 – Dec 2024

Master of Science in Robotics (Concentration: Computer Science)

GPA 3.96/4.0

National Institute of Technology Silchar, Assam, India

July 2018 – May 2022

Bachelor of Technology – Mechanical Engineering

GPA 8.98/10.0

TECHNICAL SKILLS

Programming Languages: Python, R, C++, JavaScript, Java, MATLAB, Groovy

Cloud & DevOps: Azure, AWS (EC2, Lambda, S3, Batch, Sagemaker, Bedrock, Fargate, ECS, EFS, FSX, EBS, ECR), Serverless, Docker, GitHub, GitHub Actions, GitLab, CI/CD, NGINX, Linux

Frameworks & libraries: PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas, Nextflow, Dash, Figma

Web Technologies: FastAPI, **Svelte**, React, Node.js, Express.js, Mongoose, REST API, HTML, CSS, Bootstrap, TypeScript

Databases: MySQL, MongoDB, PostgreSQL, Athena, RDS, DynamoDB, DuckDB

AI & Other Tools: **MCP**, **Claude Code**, **Cursor**, **Windsurf**, **Kiro**, Git, Tableau, VS Code, Postman, ROS

EXPERIENCE

Long Health, USA (Software Engineer)

Feb 2026 - Present

- Building AI-powered document intelligence system using AWS Textract OCR, RAG, and LLMs to extract structured patient information and generate summaries from unstructured records for physician workflows
- Engineered LLM prompt pipeline processing **500+** patient records/day, cutting manual physician review time by more than **90%** and reducing structured data errors by **~30%**
- Wrote unit tests to validate LLM prompt engineering workflows, validating structured data extraction accuracy

UniBio Intelligence, USA (Software Engineer)

July 2025 – Feb 2026

- Designed and built scalable **FastAPI** backend services on Azure, enabling containerized computational workflows that supported concurrent multi-tenant research workloads
- Migrated legacy OpenAPI workflows to Model Context Protocol (**MCP**), modernizing service interfaces and enabling LLM-native workflow execution at scale
- Deployed containerized services via **CI/CD** pipelines cutting release cycle time and eliminating environment-specific deployment failures

Ampersand Biomedicines, Boston, USA (Machine Learning Engineer Co-op)

Feb 2025 – July 2025

- Operated and automated cloud-based compute pipelines for large-scale biological data processing
- Built containerized AWS Batch workloads for long-running genomics jobs, improving fault tolerance
- Eliminated recurring manual provisioning steps using Python and AWS tooling across large-scale biological data processing jobs
- Collaborated cross-functionally with bioinformatics researchers and engineers to scope priorities and deliver milestones

Ampersand Biomedicines, Boston, USA (AI/Data Engineering Co-op)

Jan 2024 - Aug 2024

- Architected **cloud-native**, scalable data pipelines using Docker, AWS (S3, Batch, Lambda), and workflow automation tools to support reproducible large-scale biological data processing
- Optimized distributed processing jobs, cutting reduce memory usage and operational cost by **50%**

PROJECTS

API Development Using Python (FastAPI, PostgreSQL, Docker, NGINX, JWT, GitHub Actions, Ubuntu)

July 2023

- Built a production-grade **REST API** using FastAPI and PostgreSQL, integrating JWT authentication and role-based access
- Implemented **CI/CD** pipelines using **GitHub Actions** for automated build, test, and deployment workflows
- Containerized application using Docker and configured **NGINX** reverse proxy for deployment stability
- Deployed on **Linux**-based systems with configuration-driven environments

ACHIEVEMENTS & CONTRIBUTIONS

- Co-authored and presented research on **A Fish Robot: It's Modeling and Pose Estimation** at the [2021 International Symposium of Asian Control Association on Intelligent Robotics and Industrial Automation \(IRIA\)](#)
- **Teaching Assistant** for **Programming with Data** course assisting students with assignments and labs across **2 semesters**
- Recipient of '**Assistance to Meritorious Students Scholarship – Junior Level**' for outstanding academic performance

COURSES & CERTIFICATIONS

AWS – AI Practitioner | Algorithms | Neural Networks and Deep Learning | Natural Language Processing | Pattern Recognition and Computer Vision | Human Computer Interaction